

Bachelor Thesis Topics Descriptions

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Theme A: Reasoning-Augmented Large Language Models

Large Language Models (LLMs) can generate fluent and convincing text, yet they often struggle with reliable reasoning, consistent decision-making, and providing trustworthy explanations. This theme focuses on knowledge-augmented LLM systems, where explicit knowledge representation and reasoning techniques are integrated into LLM-based pipelines to support more reliable and explainable AI behavior. Instead of relying on the LLM alone, these systems use structured knowledge such as rules, constraints, and causal relations to guide, verify, and explain model outputs. The projects in this theme study how symbolic reasoning can be used to detect hallucinations, generate verifiable explanations, support causal reasoning, and ensure consistency in LLM-based systems.

Bachelor Thesis Topic Title:	(A1) LLM Hallucination Detection via Knowledge Augmentation
Offered to (Major):	Computer Science and Engineering (CSEN)
Short Description:	LLMs can generate fluent responses that violate basic domain knowledge or logical constraints. This project explores how explicit domain knowledge can be used to detect hallucinations in LLM outputs by checking consistency with known rules and facts. The objective of this project is to design a knowledge-grounded pipeline that identifies hallucinated or inconsistent LLM outputs using symbolic reasoning rather than model fine-tuning. Required Tasks: 1. Select a restricted domain (finance, policy, or games) with well-defined rules. 2. Formalize domain knowledge as symbolic constraints or rules. 3. Generate LLM responses for domain-specific queries. 4. Implement a reasoning component to check constraint violations. 5. Classify detected hallucinations into meaningful categories. 6. Compare hallucination rates with and without symbolic checking.

<i>Bachelor Thesis Topic Title:</i>	(A2) Verifiable Explanations Generation for LLM Reasoning
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	Although LLMs can produce explanations for their answers, these explanations are often unfaithful or logically incorrect. This project studies how explanations can be represented in a structured, symbolic form and verified using reasoning techniques. The objective is to move from fluent explanations to verifiable explanations by grounding them in explicit symbolic representations. Required Tasks: 1. Define a structured representation for explanations (e.g., rules, argument structures). 2. Design prompts that encourage structured explanation generation. 3. Implement a symbolic verifier to check explanation correctness. 4. Evaluate whether explanations logically support the conclusions. 5. Analyze explanation failure cases.

<i>Bachelor Thesis Topic Title:</i>	(A3) Causal and Counterfactual Reasoning in LLMs
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	LLMs often confuse correlation with causation and struggle with counterfactual reasoning. This project integrates explicit causal knowledge into LLM-based systems to improve causal and counterfactual reasoning. The objective is to investigate whether grounding LLMs in causal models improves their ability to answer causal and counterfactual questions. Required Tasks: 1. Select a domain with clear causal relationships. 2. Represent causal knowledge using causal graphs. 3. Design LLM prompts informed by causal structure. 4. Implement symbolic checks for causal consistency. 5. Evaluate performance on causal and counterfactual queries. 6. Compare causal-aware reasoning with baseline LLM reasoning.

<i>Bachelor Thesis Topic Title:</i>	(A4) Belief-Aware Reasoning and Belief Revision in LLMs
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	LLMs often produce inconsistent responses when presented with new or updated information, as they lack an explicit representation of beliefs and belief change. This project studies belief-aware knowledge-grounded LLM systems, where a single LLM-based agent maintains an explicit set of beliefs about the world and updates these beliefs when new information becomes available. Rather than regenerating answers from scratch, the system reasons about what has changed, revises its beliefs accordingly, and ensures consistency over time. The objective is to explore how explicit belief representation and belief revision mechanisms can improve the reliability and consistency of LLM-based reasoning. Required Tasks: 1. Define a domain where information may change over time. 2. Represent the agent's beliefs explicitly using a structured or symbolic format 3. Design LLM prompts that reason based on the current belief state 4. Introduce new or conflicting information incrementally 5. Implement belief update or belief revision rules 6. Evaluate consistency of system responses before and after belief updates

Theme B: Knowledge Representation and Reasoning in Agentic AI

Modern AI systems increasingly operate as agentic systems: they consist of multiple components or agents that collaborate, make decisions over multiple steps, and maintain internal state over time. Unlike single-step AI models, agentic systems must decide who is allowed to make decisions, how decisions are validated, how conflicts are resolved, and what happens when something goes wrong.

This theme focuses on agentic workflows, where LLM-based agents are orchestrated using tools such as AutoGen or similar frameworks. The emphasis is not on improving the reasoning ability of a single model (covered in Theme A), but on organizing, controlling, and governing reasoning at the system level. The projects study how concepts from knowledge representation and reasoning such as rules, constraints, roles, authority, consistency, and responsibility can be embedded into agentic systems to ensure reliable, transparent, and well-governed decision-making.

<i>Bachelor Thesis Topic Title:</i>	(B1) A Role-Governed Agentic Workflow for Financial Decision-Making
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	In many real financial processes, decisions are not made by a single actor. One party proposes a decision, another checks compliance, and a third approves or rejects it. This project explores role-governed agentic workflows, where different agents have clearly defined responsibilities within a financial decision-making pipeline. The focus is on decision flow and control, not financial prediction or trading. Required Tasks: 1. Define a simplified financial decision-making scenario (e.g., budgeting, eligibility, expense approval). 2. Design an agentic workflow. 3. Assign clear roles to agents (e.g., interpretation, reasoning, validation). 4. Represent basic financial rules and constraints in a structured form. 5. Integrate rule checking as explicit steps in the workflow. 6. Evaluate decision consistency across different scenarios.

<i>Bachelor Thesis Topic Title:</i>	<i>(B2) A Policy-Aware Agentic System for Compliance Reasoning</i>
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	Many organizational and administrative processes are governed by policies that include obligations, permissions, and exceptions. This project explores a policy-aware agentic system, where policy checks are explicitly embedded as steps in an agent workflow. Instead of relying on implicit prompt instructions, the system reasons about compliance during execution and detects policy violations in a transparent manner. Required Tasks: 1. Select a simplified policy or compliance domain. 2. Represent policies, obligations, and exceptions in a structured format. 3. Design an agentic workflow with explicit policy-checking stages. 4. Use LLM-based agents to reason about compliance scenarios. 5. Detect and explain policy violations during execution. 6. Compare compliance outcomes with and without explicit policy checks. 7. Analyze how exceptions affect agent decisions.

<i>Bachelor Thesis Topic Title:</i>	<i>(B3, B4, B5) An Agentic System for Reasoning in Rule-Based Games</i>
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	Rule-based games provide a controlled environment for studying reasoning and decision-making. This project designs an agentic workflow for reasoning in rule-based games, where agents propose actions, verify rule compliance, and update game state across multiple turns. The focus is on consistent rule-following and reasoning, not on optimal gameplay or performance. Candidate games include simple rule-based games such as Tic-Tac-Toe (or its variants), Connect Four, and Chess. Three students are needed to work on this project, each working on a different game. Required Tasks: 1. Select one or more rule-based games from the proposed candidates. 2. Formalize the game rules and legal move conditions.

	3. Design an agentic workflow for turn-based reasoning. 4. Assign agents roles (e.g., move proposal, rule checking, state update). 5. Implement rule compliance checks for proposed actions. 6. Detect and analyze illegal or inconsistent moves. 7. Compare LLM-only reasoning with rule-grounded agentic reasoning.
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<i>Bachelor Thesis Topic Title:</i>	(B6) Argumentation-Based Decision Making in Agentic AI Systems
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	Agentic AI systems often involve multiple agents that propose actions, provide justifications, and validate or reject decisions. However, in many existing systems, conflicts between agents' recommendations are resolved in an ad-hoc manner, and the reasons why one decision is chosen over another are not made explicit. This project explores argumentation-based decision making in agentic AI systems, where decisions are selected through structured reasoning over competing arguments. Each agent contributes arguments supporting or opposing certain actions, and an argumentation mechanism is used to resolve conflicts, determine which arguments prevail, and justify the final decision. Rather than focusing on improving individual agent reasoning, the project studies how argumentation frameworks can serve as a governance layer inside agentic workflows, enabling transparent conflict resolution, structured justification, and traceable decision outcomes. Required Tasks: 1. Design an agentic workflow in which multiple agents propose actions and corresponding justifications. 2. Define a structured representation of arguments, including claims and their supporting reasons. 3. Model conflicts between arguments (e.g., attacks between opposing justifications). 4. Implement an argumentation evaluation mechanism to determine which arguments are accepted. 5. Use the accepted arguments to select a final system decision. 6. Generate structured explanations based on the accepted arguments and rejected alternatives.

<i>Bachelor Thesis Topic Title:</i>	(B7) Communication and Cooperation in LLM-Based Agentic Systems
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	Multi-agent systems (MAS) rely fundamentally on communication and cooperation mechanisms to coordinate actions, share information, and achieve collective goals. With the emergence of Large Language Models (LLMs), recent research has revisited classical MAS communication paradigms using natural-language interaction, shared memory, and hybrid neuro-symbolic reasoning. These systems span a range of organizational structures, including centralized coordination via hub agents, hierarchical orchestration layers, and decentralized peer-to-peer interaction. This project focuses on reviewing and analyzing key survey and review studies on communication and cooperation in LLM-based multi-agent systems, with particular attention to centralized and decentralized coordination structures. The reviewed literature covers LLM-based MAS applied to software engineering, autonomous agents, collective problem solving, and memory-driven cooperation, as well as neuro-symbolic multi-agent architectures that combine LLMs with symbolic planning and reasoning components. Classical (non-LLM-specific) surveys on cooperative multi-agent communication are also examined to provide foundational communication models that remain applicable to symbolic and neuro-symbolic agentic systems. Required Tasks: 1. Identify and review key survey and review papers on communication and cooperation in multi-agent systems, including both classical MAS and LLM-based MAS. 2. Analyze coordination structures in LLM-based multi-agent systems, focusing on centralized, hierarchical, and decentralized approaches. 3. Compare LLM-based communication mechanisms with classical MAS communication models. 4. Summarize major application areas discussed in the literature. 5. Synthesize the reviewed work into a clear taxonomy of communication and cooperation mechanisms. 6. Identify open challenges and research gaps highlighted by existing surveys.

<i>Bachelor Thesis Topic Title:</i>	<i>(B8) Responsibility and Accountability in Agentic AI Systems</i>
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	As agentic AI systems become more autonomous, an important question arises: who is responsible when something goes wrong? In systems composed of multiple interacting components, responsibility is often unclear. This project is research-focused and studies responsibility and accountability in agentic AI systems by analyzing system design, decision traces, and role assignments. Required Tasks: 1. Review recent agentic AI systems and identify accountability challenges. 2. Define roles and decision responsibilities within agentic workflows. 3. Propose a structured representation of responsibility and authority. 4. Design example workflows illustrating responsibility assignment. 5. Analyze failure cases and responsibility attribution. 6. Compare implicit vs explicit responsibility models. 7. Discuss implications for trust, safety, and AI governance.

<i>Bachelor Thesis Topic</i>	<i>(B9) A Review and Critique of Evaluation Metrics for Agentic AI</i>
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<i>Title:</i>	Systems
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	Agentic AI systems are increasingly used to perform complex tasks involving multiple steps, interacting agents, memory, and decision-making over time. While many such systems have been proposed, evaluating their performance remains an open challenge. Existing evaluation approaches often rely on task success, accuracy, or subjective judgments, which may not adequately capture properties such as reasoning quality, consistency, robustness, coordination, or accountability. This project focuses on reviewing and critically analyzing evaluation metrics used for agentic AI systems, particularly those based on Large Language Models. The goal is to understand what is currently being measured, what is overlooked, and how evaluation practices align (or fail to align) with the goals of agentic AI, such as reliable decision-making, governance, and explainability. The project is literature-based and emphasizes analysis, comparison, and critique rather than implementation. Required Tasks: 1. Identify and review key survey and research papers on agentic AI systems, focusing on how they are evaluated. 2. Collect and categorize evaluation metrics used in agentic AI literature (e.g., task success, consistency, reasoning quality, robustness, coordination effectiveness, human oversight). 3. Analyze how different metrics relate to system-level properties, such as decision governance, reliability, and accountability. 4. Compare evaluation practices across different types of agentic systems (e.g., centralized vs decentralized, single-agent workflows vs multi-agent systems). 5. Critically assess limitations and blind spots of current evaluation approaches. 6. Synthesize findings into a structured comparison or taxonomy of evaluation metrics. 7. Identify open challenges and directions for evaluating agentic AI systems more meaningfully.

Autonomous vehicles operate in environments where decisions have legal, safety, and ethical consequences. Unlike many AI systems, incorrect decisions may lead to irreversible harm, making reliable reasoning and decision-making critical. While much autonomous driving research focuses on perception and control, this theme emphasizes high-level reasoning under formal rules, safety constraints, and interaction with other vehicles.

The projects in this theme study how knowledge representation, reasoning, and agentic decision pipelines can support safe, lawful, and explainable driving behavior, especially in situations where rules, safety goals, and interactions may conflict. All topics abstract away from raw sensor processing and vehicle dynamics, focusing instead on decision logic, rule interpretation, safety prioritization, and accountability.

<i>Bachelor Thesis Topic Title:</i>	<i>(C1) Norm-Constrained and Safety-Critical Decision Making for Autonomous Vehicles</i>
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	Autonomous vehicles must make driving decisions that satisfy multiple constraints, including traffic laws, safety requirements, and situational norms. In many driving scenarios, these constraints restrict the available actions but still allow at least one legally and safely permissible option. This project focuses on norm-constrained decision-making, where an autonomous vehicle must reason about which actions are admissible under a combination of legal and safety constraints. Required Tasks: 1. Define simplified driving scenarios (e.g., intersections, lane changes, merging). 2. Formalize traffic rules in a structured or symbolic representation. 3. Use an LLM to propose driving actions for given scenarios. 4. Implement rule-checking mechanisms to validate proposed actions. 5. Detect and analyze rule violations.



<i>Bachelor Thesis Topic Title:</i>	<i>(C2) Legal versus Safe Behavior: Reasoning About Exceptional Driving Situations for Autonomous Vehicles</i>
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	Autonomous vehicles are expected to follow traffic laws while ensuring safety. However, real-world driving often involves exceptional situations where strictly legal behavior may be unsafe, or where safe behavior may violate traffic rules. Human drivers routinely resolve such conflicts using judgment and contextual reasoning, but autonomous systems lack an explicit mechanism to reason about these trade-offs. This project studies legal versus safe behavior in autonomous driving, focusing on how an autonomous vehicle can reason when no available action fully satisfies both legal and safety constraints. Instead of assuming that rules can always be followed, the project explores norm conflicts, exceptional decision-making, and justified rule violations. The goal is to design a reasoning-based decision framework that classifies actions as permissible, impermissible, or exceptional, and generates structured justifications for decisions made under conflict. The project abstracts away from perception and vehicle control, focusing on high-level decision logic, norm prioritization, and explainability in safety-critical scenarios. Required Tasks 1. Identify exceptional driving scenarios where no action fully satisfies both legal and safety constraints. 2. Represent relevant legal norms and safety constraints in a structured form. 3. Define candidate driving actions and detect norm conflicts. 4. Design a reasoning mechanism that selects the least unacceptable action, prioritizing safety. 5. Generate structured justifications explaining rule violations and rejected alternatives. 6. Compare system behavior with rule-only and safety-only baselines.



<i>Bachelor Thesis Topic Title:</i>	(C3) Multi-Vehicle Interaction and Coordination Reasoning
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	Real-world traffic involves interactions between multiple vehicles that must coordinate implicitly to avoid conflicts. This project focuses on reasoning about interactions between autonomous vehicles, modeled as decision-making agents operating under uncertainty. The emphasis is on anticipating other vehicles' actions, reasoning about right-of-way and yielding behavior, and handling situations where coordination is required but communication is limited or absent. The project studies how interaction reasoning affects safety and conflict avoidance. Required Tasks: 1. Define traffic scenarios involving multiple autonomous vehicles. 2. Model each vehicle as a reasoning agent with its own decisions. 3. Represent interaction rules and constraints (e.g., priority, yielding). 4. Simulate interactions and detect conflicts. 5. Evaluate coordination success and failure cases.



<i>Bachelor Thesis Topic Title:</i>	<i>(C4) A Framework for Explainable Autonomous Driving Decisions</i>
<i>Offered to (Major):</i>	Computer Science and Engineering (CSEN)
<i>Short Description:</i>	For autonomous vehicles to be trusted by users, regulators, and society, their decisions must be explainable. This project focuses on developing a framework for explainable autonomous driving decisions, where driving actions are accompanied by structured, reasoning-based explanations grounded in traffic rules and safety constraints. Unlike generic explainability methods, the explanations in this project are intended to support accountability and post-incident analysis, answering questions such as <i>why a specific action was taken and why alternatives were rejected</i>. Required Tasks: 1. Select driving scenarios requiring non-trivial decisions. 2. Define a structured representation for decision explanations. 3. Generate explanations for driving actions. 4. Verify explanations against traffic rules and constraints. 5. Analyze explanation correctness and completeness. 6. Compare free-text explanations with structured explanations.

This project focuses on building a reasoning system for an artificial mind, where reasoning is performed over multiple mental attitudes, not just knowledge. The system, called MindGRAF (Mind Graphical Reasoner with Attitude Frames), represents an artificial mind as a structured graph that captures different mental attitudes like beliefs, goals, intentions, and norms, as well as the relationships between them.

MindGRAF extends the idea of knowledge graphs by allowing different types of mental attitudes to coexist and influence each other. For example, a belief may support or contradict a goal, an obligation may restrict an intention, or a new belief may require revising previous intentions. Reasoning in MindGRAF consists of updating and deriving new mental attitudes based on the current state of the artificial mind.

The project emphasizes practical system design and implementation, focusing on building a working reasoning engine, designing clear data structures, and applying good software engineering practices. The goal is not to simulate human psychology, but to engineer a structured reasoning system inspired by how humans think.

<i>Bachelor Thesis Topic Title:</i>	(D1) Inference in MindGRAF
<i>Offered to (Major):</i>	<i>Computer Science and Engineering (CSEN)</i>
<i>Short Description:</i>	Reasoning in AI systems is often limited to deriving new facts from existing knowledge. This project explores inference in the presence of multiple mental attitudes, where conclusions may depend on beliefs, goals, intentions, and obligations at the same time. The project studies how an artificial mind can draw conclusions such as <i>what should be done</i>, <i>what is allowed</i>, or <i>what is intended</i>, based on interactions between different attitudes. The focus is on designing and implementing inference rules that operate over multiple attitude types and understanding how these rules interact. Required Tasks: 1. Define a set of inference rules involving multiple mental attitudes. 2. Implement inference using such rules in the existing MindGRAF system. 3. Demonstrate the implemented inference on example decision-making scenarios involving multiple attitudes. 4. Write a test suite for the inference package.



<i>Bachelor Thesis Topic Title:</i>	(D2) Mental Attitudes Revision in MindGRAF
<i>Offered to (Major):</i>	<i>Computer Science and Engineering (CSEN)</i>
<i>Short Description:</i>	In classical AI, belief revision studies how a system should update its beliefs when new information contradicts existing knowledge. However, real reasoning involves multiple mental attitudes, such as beliefs, goals, intentions, and obligations, which may also conflict and require revision. This project studies multi-attitude revision, extending the idea of belief revision to settings where several types of attitudes coexist. The project focuses on how an artificial mind should revise its internal state when new information affects not only beliefs, but also goals or obligations. For example, a new belief may invalidate an intention, or a new obligation may override an existing goal. The emphasis is on designing clear revision principles and implementing them in the MindGRAF framework. Required Tasks: 1. Design how the supports can be defined for propositions in the different attitudes. 2. Implement a mechanism to define and detect conflicts. 3. Implement automatic conflict resolution. 4. Write a test suite demonstrating different types and conflicts and validating that they are handled correctly.



<i>Bachelor Thesis Topic Title:</i>	(D3) An LLM-Powered User Interface for MindGRAF
<i>Offered to (Major):</i>	<i>Computer Science and Engineering (CSEN)</i>
<i>Short Description:</i>	Graph-based reasoning systems are powerful but often difficult for users to interact with directly. This project focuses on building an LLM-powered user interface that allows users to interact with MindGRAF using natural language. The system translates English statements into structured graph representations that can be processed by the MindGRAF reasoning and revision engines. The project emphasizes bridging natural language and structured reasoning, enabling users to add beliefs, goals, or obligations in plain English and observe how the artificial mind updates and reasons over them. Required Tasks: 1. Define a controlled natural language format for expressing mental attitudes. 2. Design prompts for translating English input into graph structures. 3. Implement an interface that converts user input into MindGRAF nodes and relations. 4. Integrate the interface with existing inference and revision components. 5. Support basic user queries about the current state of the artificial mind. 6. Demonstrate the system on example interaction sessions. 7. Analyze translation errors and limitations of LLM-based parsing.